



Processor Design

Basic Computer Architecture

Outline

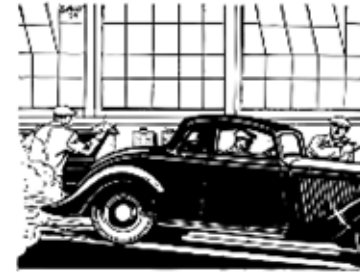
- * Overview of a Processor
- * Detailed Design of each Stage
- * The Control Unit
- * Microprogrammed Processor
- * Microassembly Language
- * The Microcontrol Unit



Processor Design

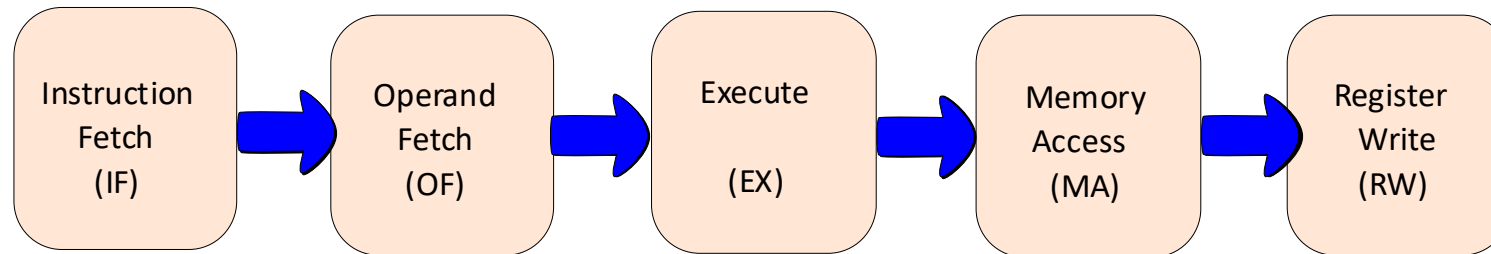
- * The aim of **processor design**
 - * **Implement** the entire SimpleRisc ISA
 - * **Process** the binary format of instructions
 - * Provide as much of **performance** as possible
- * Basic Approach
 - * Divide the processing into **stages**
 - * Design each **stage** separately

A Car Assembly Line



- * Similar to a car **assembly line**
 - * Cast raw **metal** into the **chassis** of a car
 - * Build the **Engine**
 - * Assemble the **engine** and the **chassis**
 - * Place the **dashboard**, and **upholstery**

A Processor Divided Into Stages



* Instruction Fetch (IF)

- * Fetch an instruction from the instruction memory
- * Compute the address of the next instruction

Operand Fetch (OF) Stage

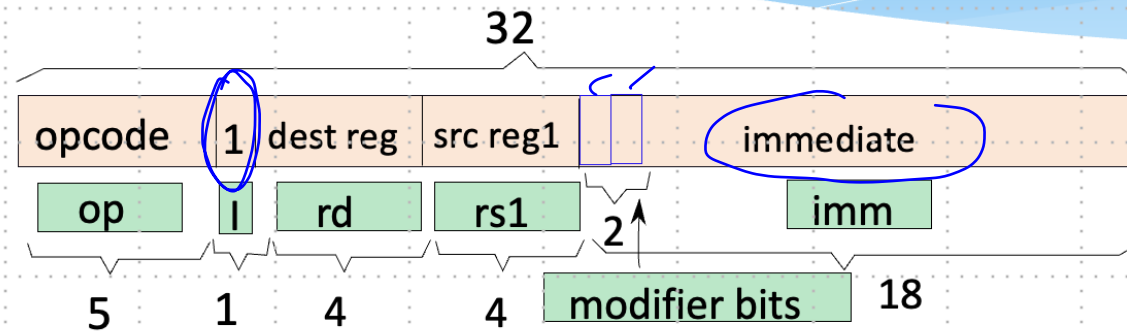
* Operand Fetch (OF)

- * **Decode** the instruction (break it into fields)
- * **Fetch** the register operands from the register file
- * **Compute** the **branch target** (PC + offset) ✓
- * **Compute** the **immediate** (16 bits + 2 bit modifiers)
- * **Generate control signals** (we will see later)

Control unit

Recap from SimpleRISC ISA

Immediate Format



- * opcode → type of the instruction
- * **I** bit → 1 (second operand is an immediate)
- * dest reg → rd
- * **source register 1** → rs1
- * **Immediate** → imm
- * **modifier** bits → 00 (default), 01 (u), 10 (h)

Execute (EX) Stage

* The EX Stage

- * Contains an **Arithmetic-Logical Unit (ALU)**

- * This unit can perform all **arithmetic** operations (add, sub, mul, div, cmp, mod), and **logical** operations (and, or, not)

- * Contains the **branch** unit for computing the branch condition (beq, bgt)

- * Contains the **flags** register (updated by the cmp instruction)

Special **flags** register → contains the result of the last comparison

- * **flags.E** = 1 (equality), **flags.GT** = 1 (greater than)

MA and RW Stages

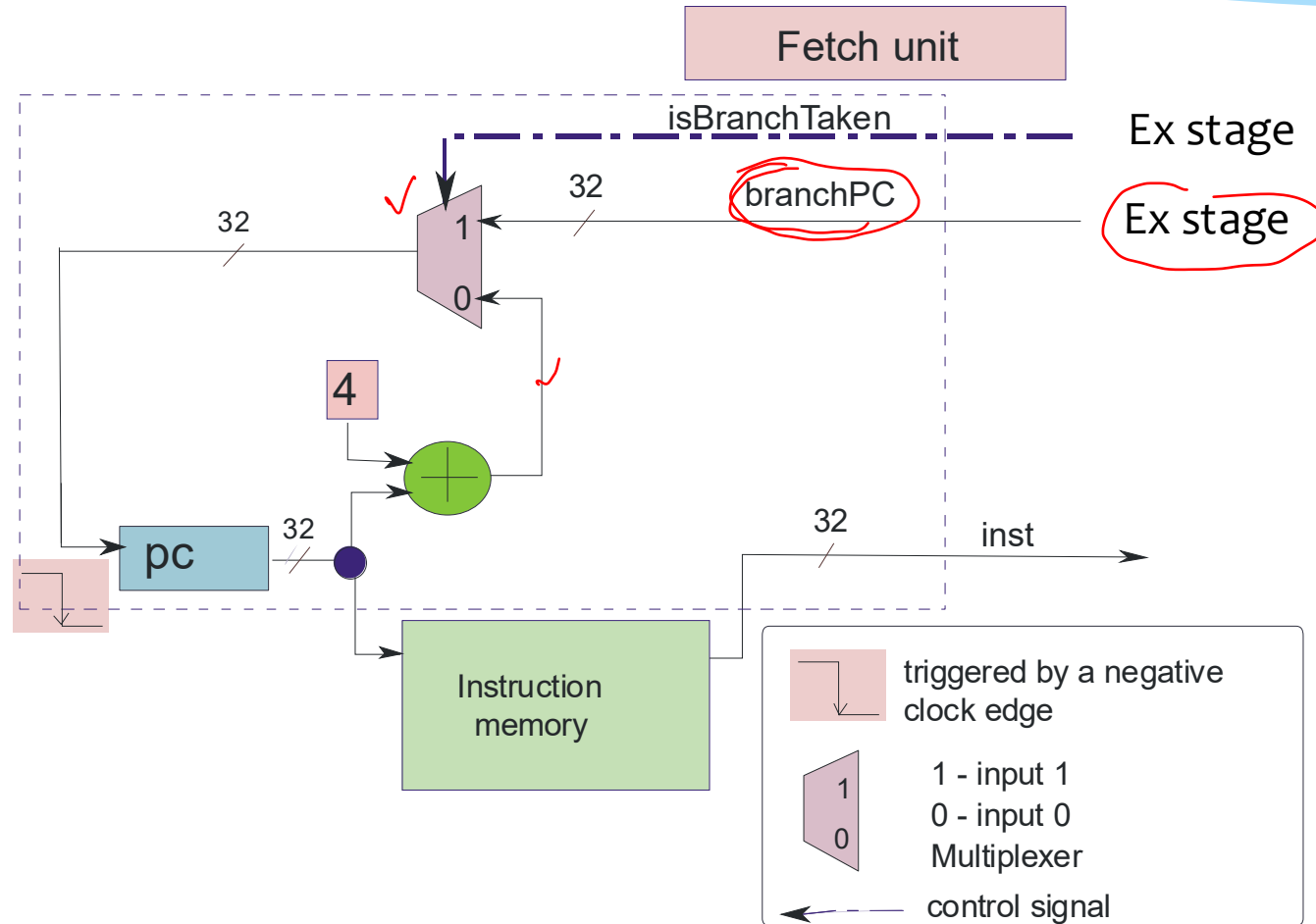
- * MA (Memory Access) Stage
 - * **Interfaces** with the **memory system**
 - * **Executes** a load or a store
- * RW (Register Write) Stage
 - * **Writes** to the register file
 - * In the case of a **call** instruction, it **writes** the return address to register, **ra**

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Instruction Fetch (IF) Stage



The Fetch unit

- * The **pc** register contains the **program counter** (**negative edge** triggered)
- * We use the **pc** to access the instruction memory
- * The **multiplexer** chooses between
 - * $pc + 4$
 - * `branchTarget`
- * It uses a control signal $\rightarrow$ **isBranchTaken**

isBranchTaken

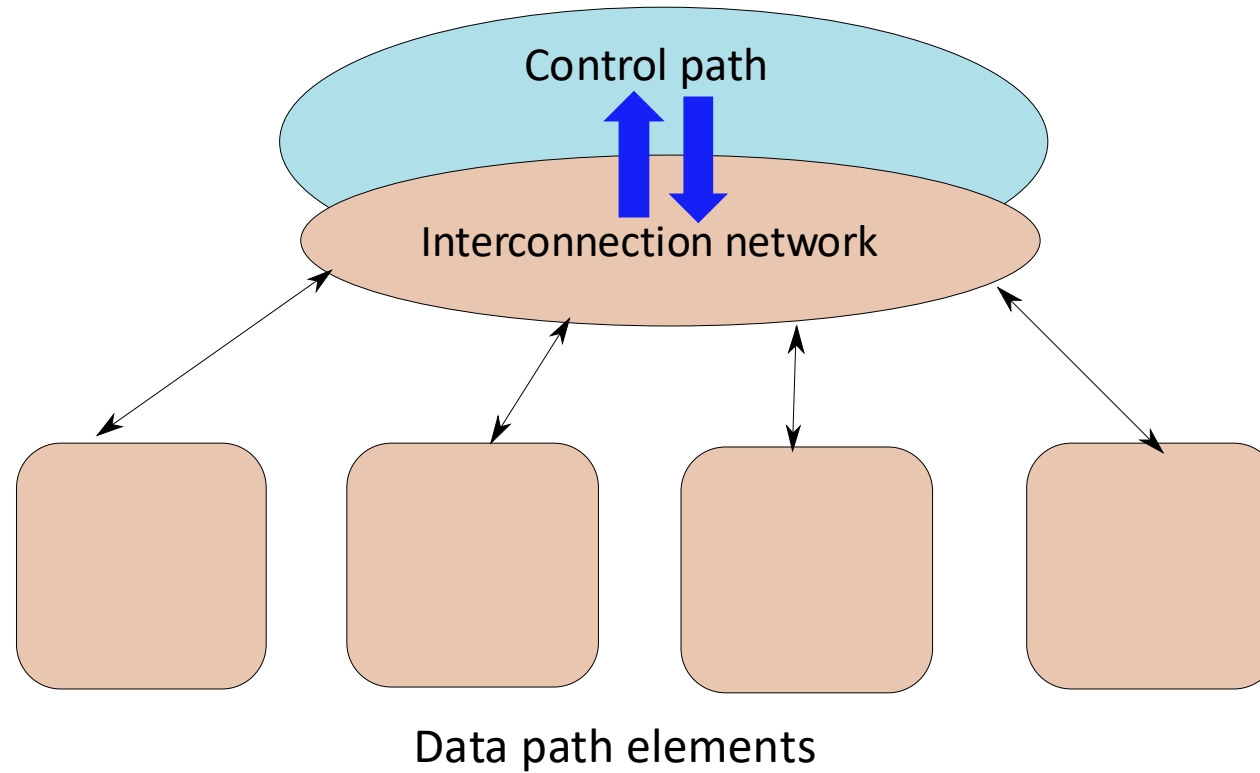
- * isBranchTaken is a **control** signal
 - * It is generated by the EX unit
- * Conditions on isBranchTaken

Instruction	Value of <i>isBranchTaken</i>
non-branch instruction	0
<i>call</i>	1
<i>ret</i>	1
<i>b</i>	1
<i>beq</i>	branch taken – 1 branch not taken – 0
<i>bgt</i>	branch taken – 1 branch not taken – 0

Data Path and Control Path

- * The **data path** consists of all the elements in a processor that are dedicated to storing, retrieving, and processing data such as **register files, memory, and the ALU**.
- * The **control path** primarily contains the **control unit**, whose role is to generate the appropriate signals to control the **movement of instructions**, and **data** in the data path.

Control Path



We will currently look at the hardwired control path.

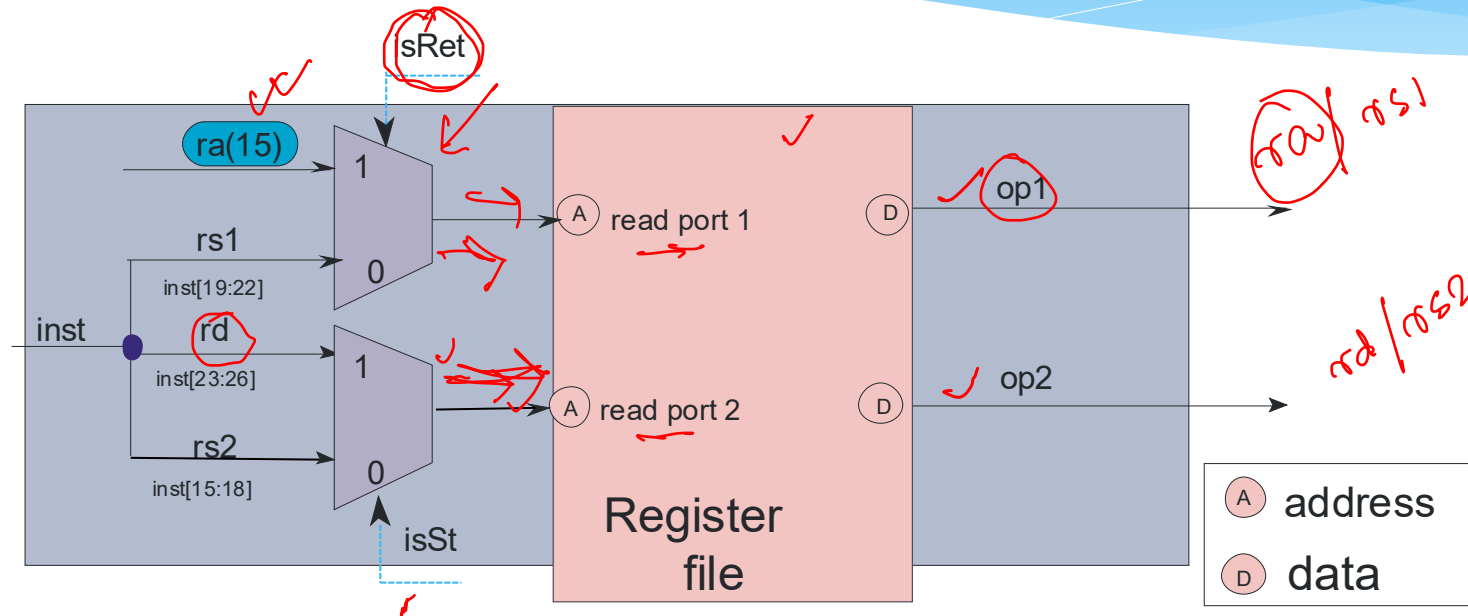
Operand Fetch Unit

Inst.	Code	Format	Inst.	Code	Format
add	00000	add rd, rs1, (rs2/imm)	lsl	01010	lsl rd, rs1, (rs2/imm)
sub	00001	sub rd, rs1, (rs2/imm)	lsr	01011	lsr rd, rs1, (rs2/imm)
mul	00010	mul rd, rs1, (rs2/imm)	asr	01100	asr rd, rs1, (rs2/imm)
div	00011	div rd, rs1, (rs2/imm)	nop	01101	nop
mod	00100	mod rd, rs1, (rs2/imm)	ld	01110	ld rd, imm[rs1]
cmp	00101	cmprs1, (rs2/imm)	st	01111	st rd, imm[rs1]
and	00110	and rd, rs1, (rs2/imm)	beq	10000	beq offset
or	00111	or rd, rs1, (rs2/imm)	bgt	10001	bgt offset
not	01000	not rd, (rs2/imm)	b	10010	b offset
mov	01001	mov rd, (rs2/imm)	call	10011	call offset
			ret	10100	ret

Instruction Formats

Format	Definition					
<i>branch</i>	<i>op</i> (28-32)	<i>offset</i> (1-27)	Word offset			
<i>register</i>	<i>op</i> (28-32)	<i>I</i> (27)	<i>rd</i> (23-26)	<i>rs1</i> (19-22)	<i>rs2</i> (15-18)	
<i>immediate</i>	<i>op</i> (28-32)	<i>I</i> (27)	<i>rd</i> (23-26)	<i>rs1</i> (19-22)	<i>imm</i> (1-18)	
<i>op</i> → opcode, <i>offset</i> → branch offset, <i>I</i> → immediate bit, <i>rd</i> → destination register						
<i>rs1</i> → source register 1, <i>rs2</i> → source register 2, <i>imm</i> → immediate operand						

Register File Read

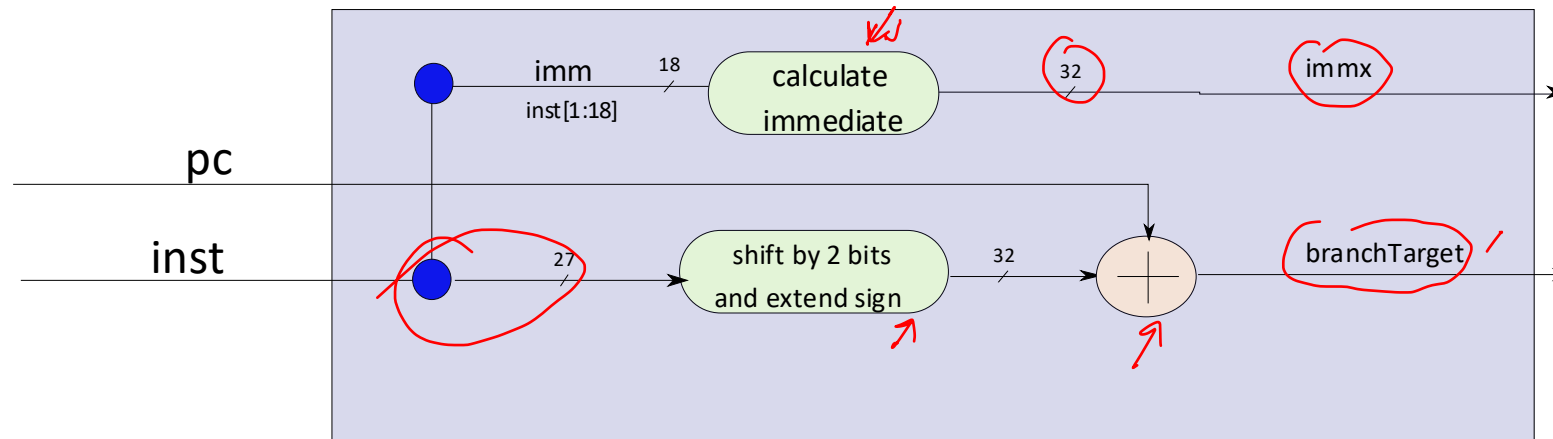


- * **First input** → `rs1` or `ra(15)` (ret instruction)
- * **Second input** → `rs2` or `rd` (store instruction)

Register File Access

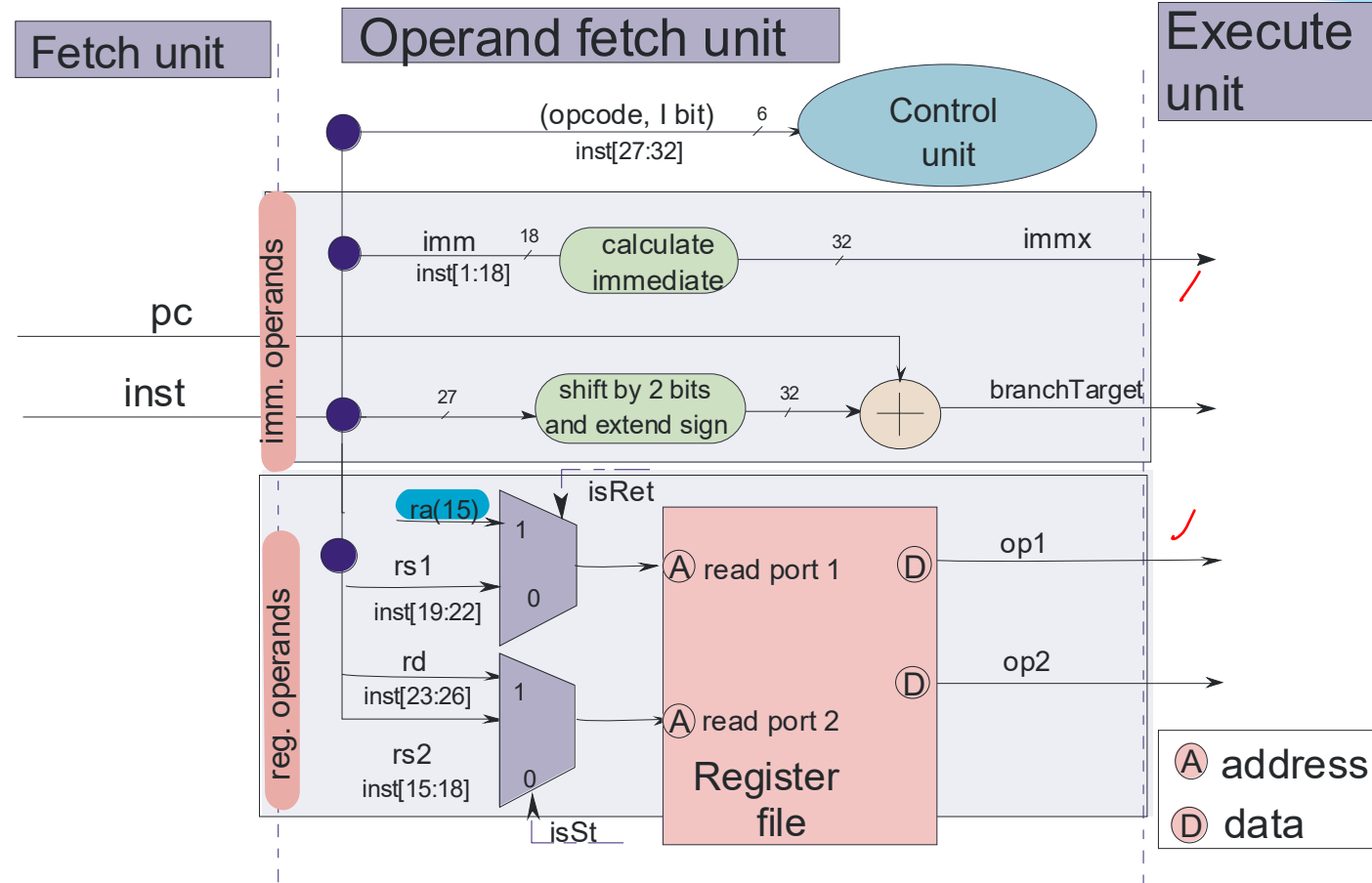
- * The **register file** has two **read ports**
 - * 1st Input
 - * 2nd Input
- * The two outputs are op1, and op2
 - * **op1** is the **branch target (return address)** in the case of a **ret** instruction, or **rs1**
 - * **op2** is the value that needs to be stored in the case of a **store** instruction, or **rs2**

Immediate and Branch Unit

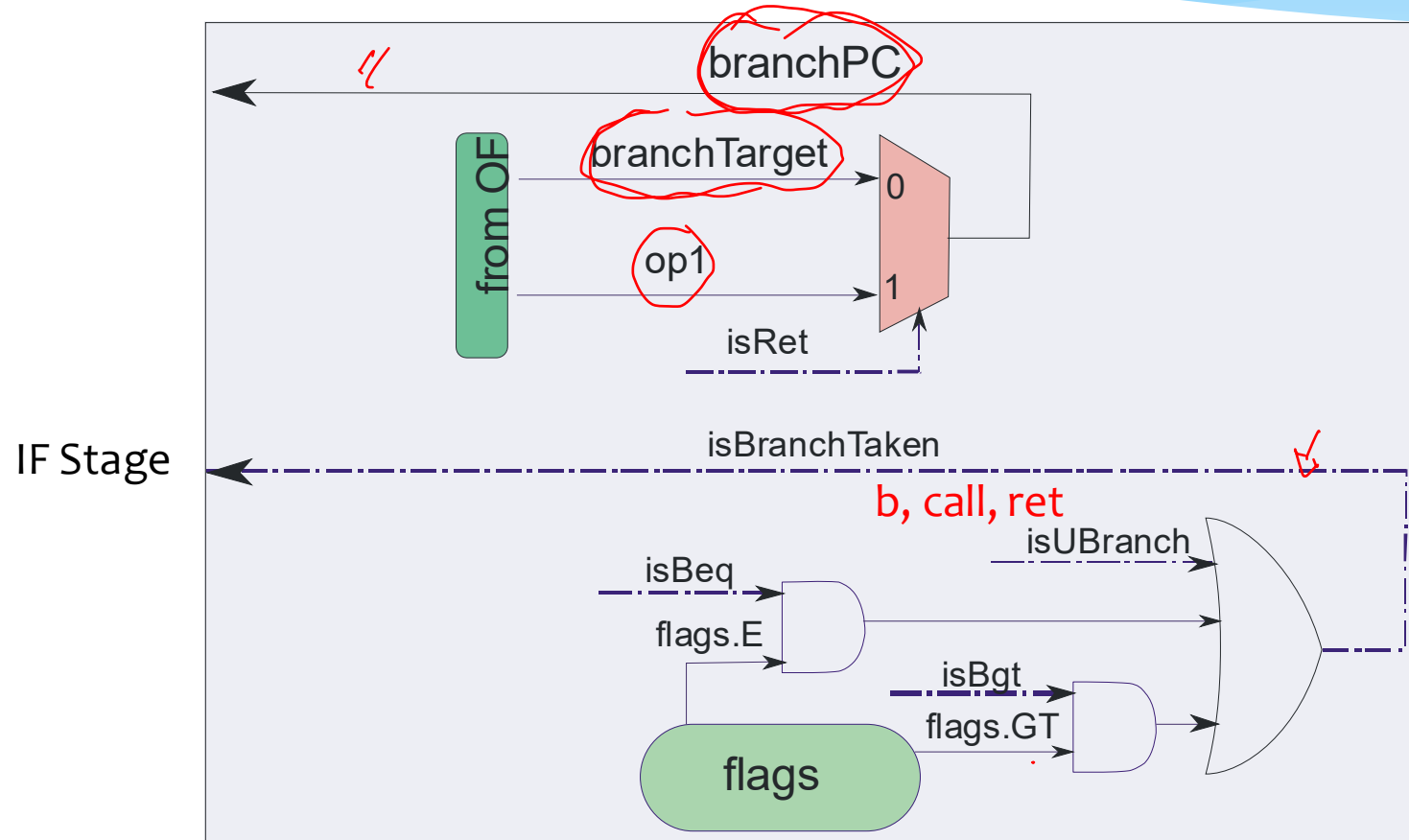


- * Compute **immx** (extended immediate), **branchTarget**, irrespective of the **instruction format**.
- * For the **branchTarget** we need to choose between the embedded target and op1 (ret)

OF Unit

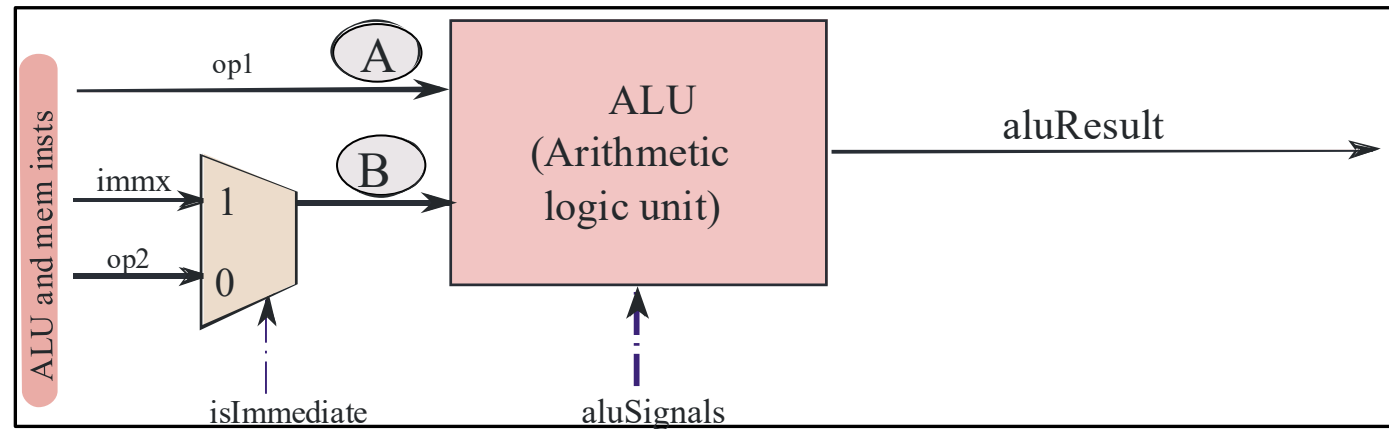


EX Stage – Branch Unit



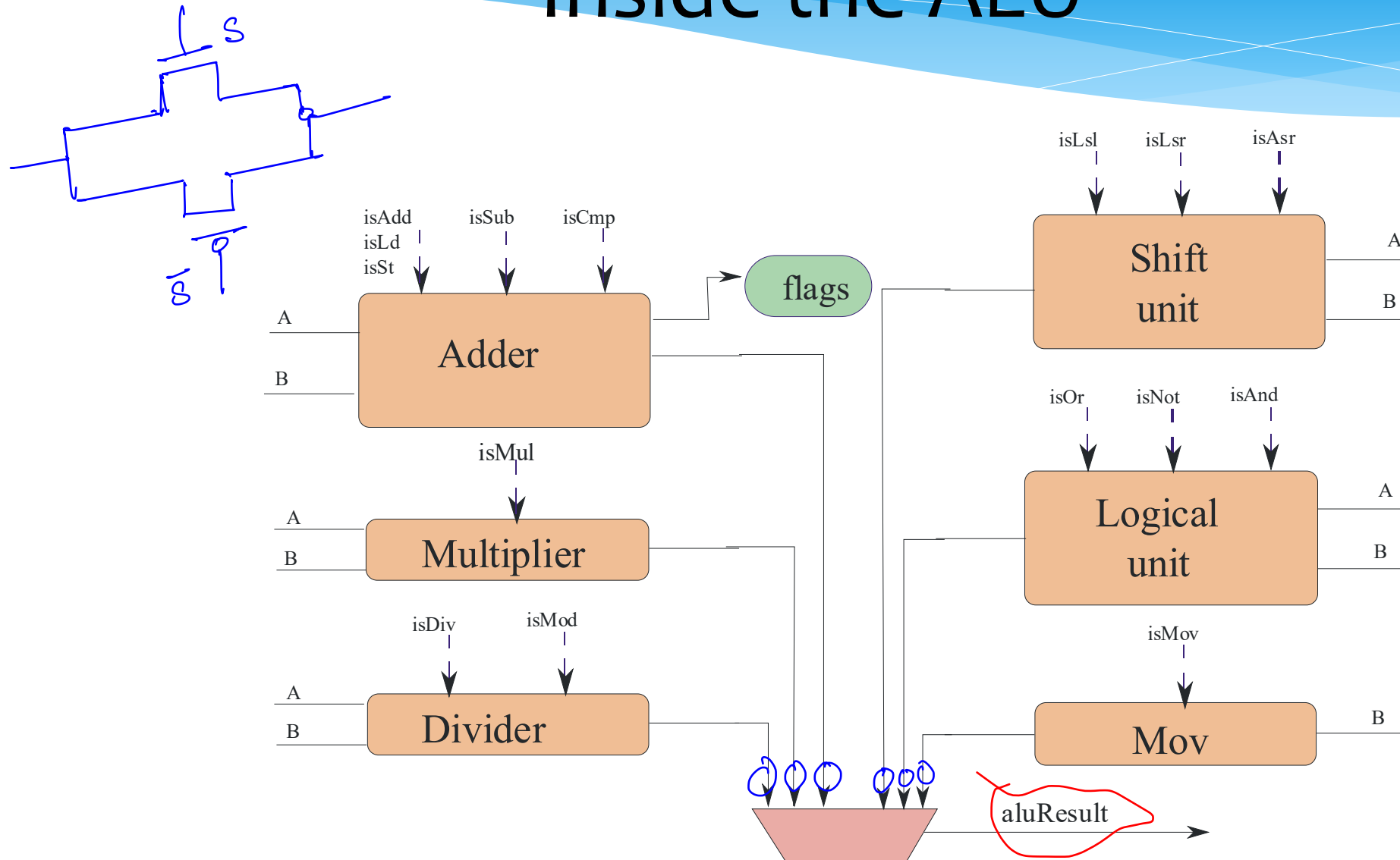
Generates the isBranchTaken Signal

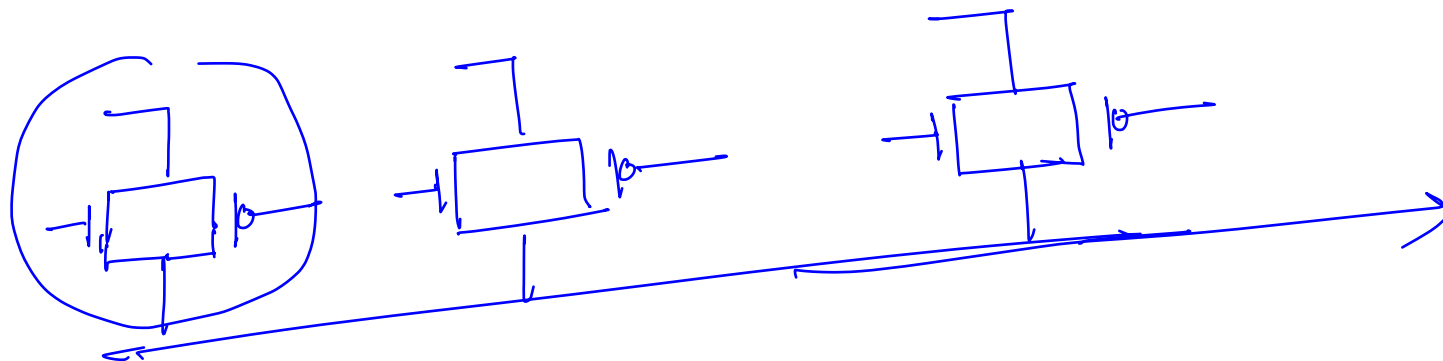
ALU ✓



Choose between immx and op2 based on the value of the I bit

Inside the ALU

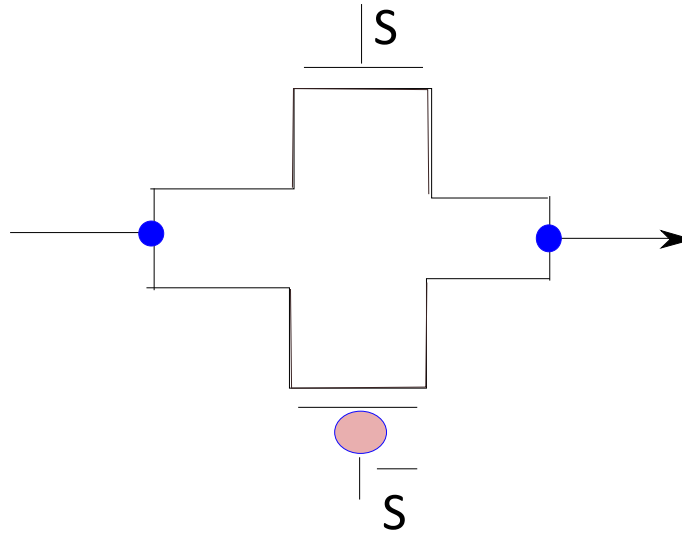




Disabling some Inputs

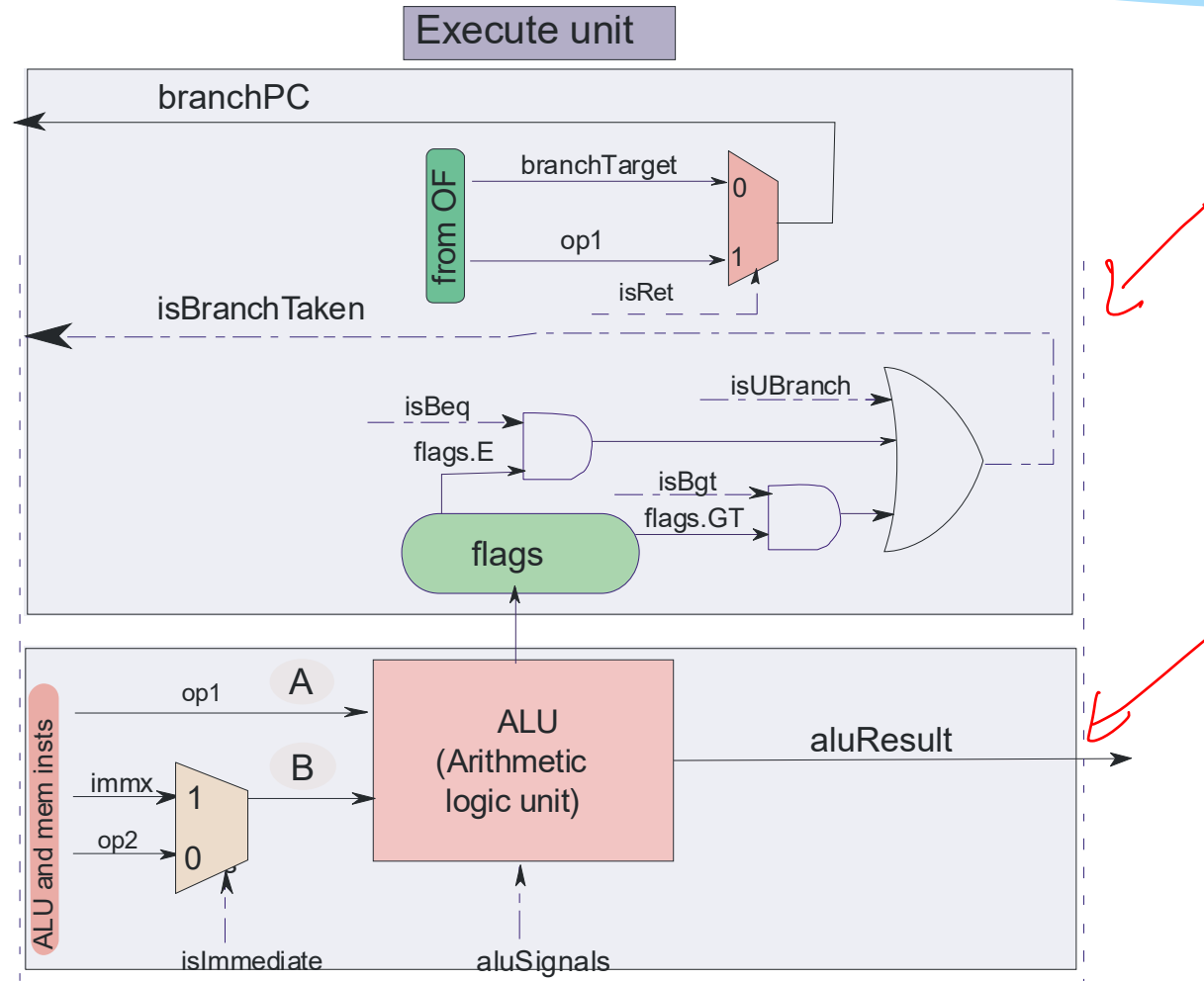
- * We do not want all the **units** of the ALU to be **active** at the same time because of we want to save **power**
- * The **instruction** will only use 1 **unit**
- * **Power** is dissipated when the **inputs** or **outputs** make a **transition** ($0 \rightarrow 1$, $1 \rightarrow 0$)
 - * We shall avoid a **transition** by not letting the new **inputs** to propagate to **units** that do not require them
 - * They will thus have the **old inputs** (**no** switching)

Use a Transmission Gate

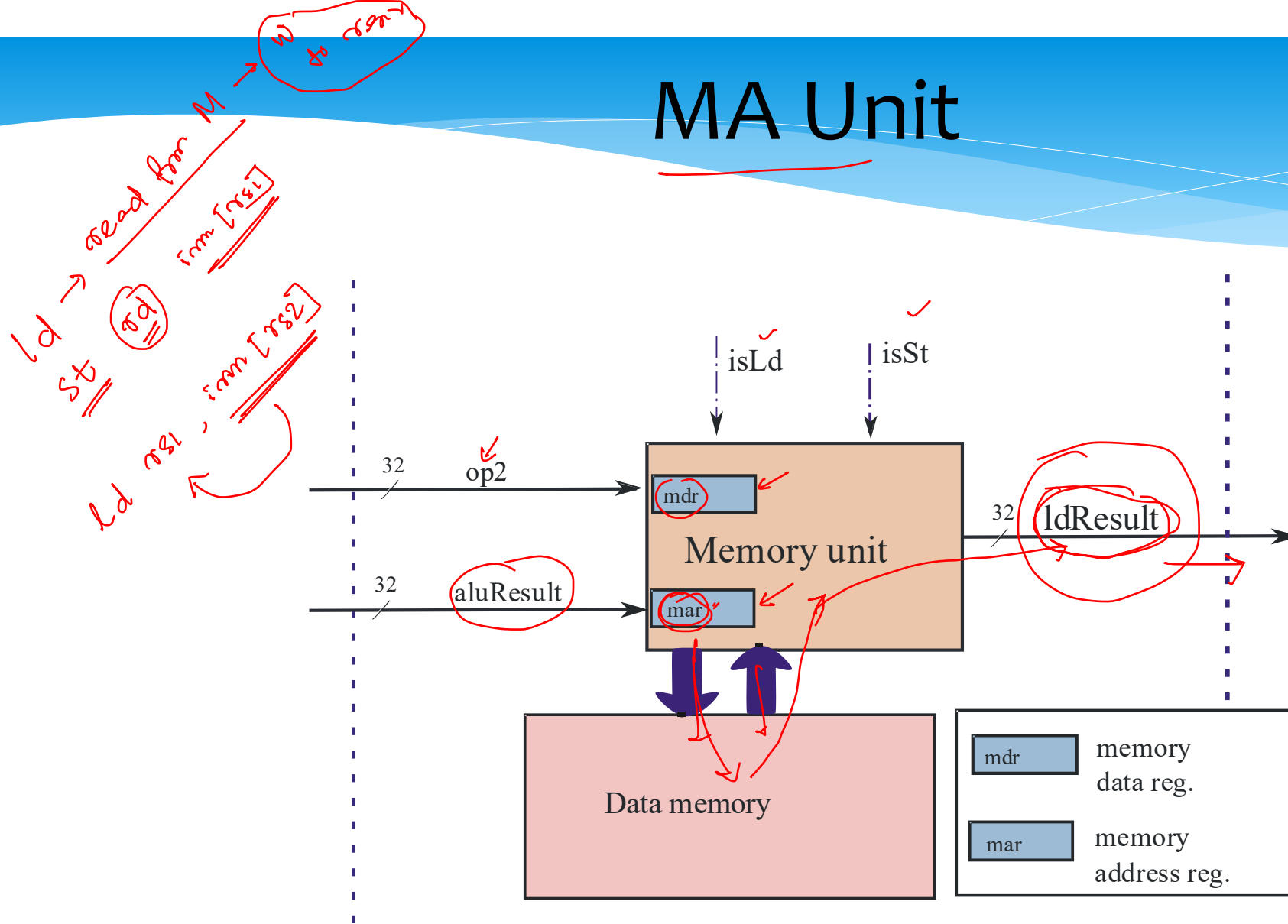


- * output = input (if $S = 1$)
- * Otherwise, the **output** is totally disconnected from the **input**

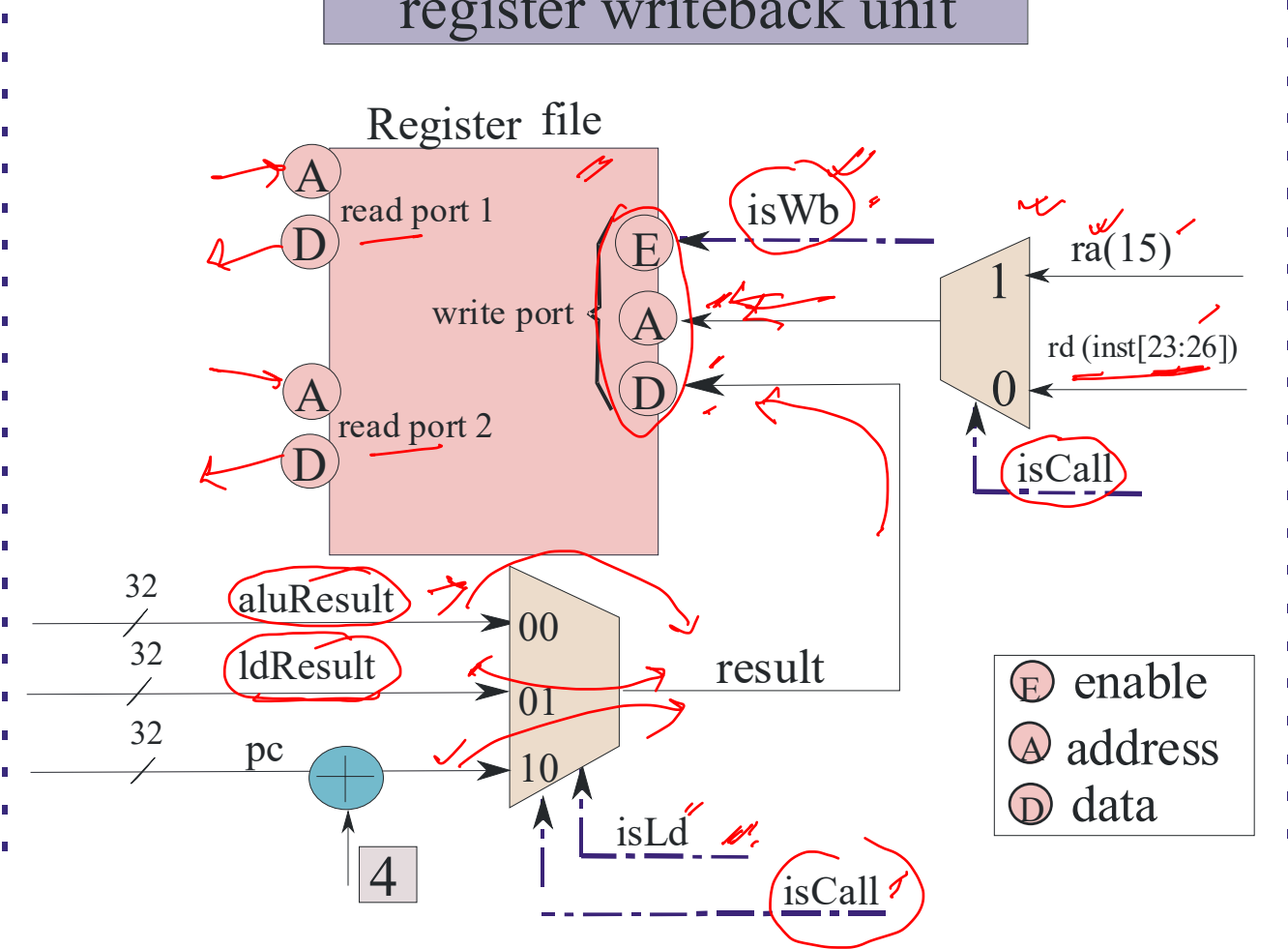
EX Unit

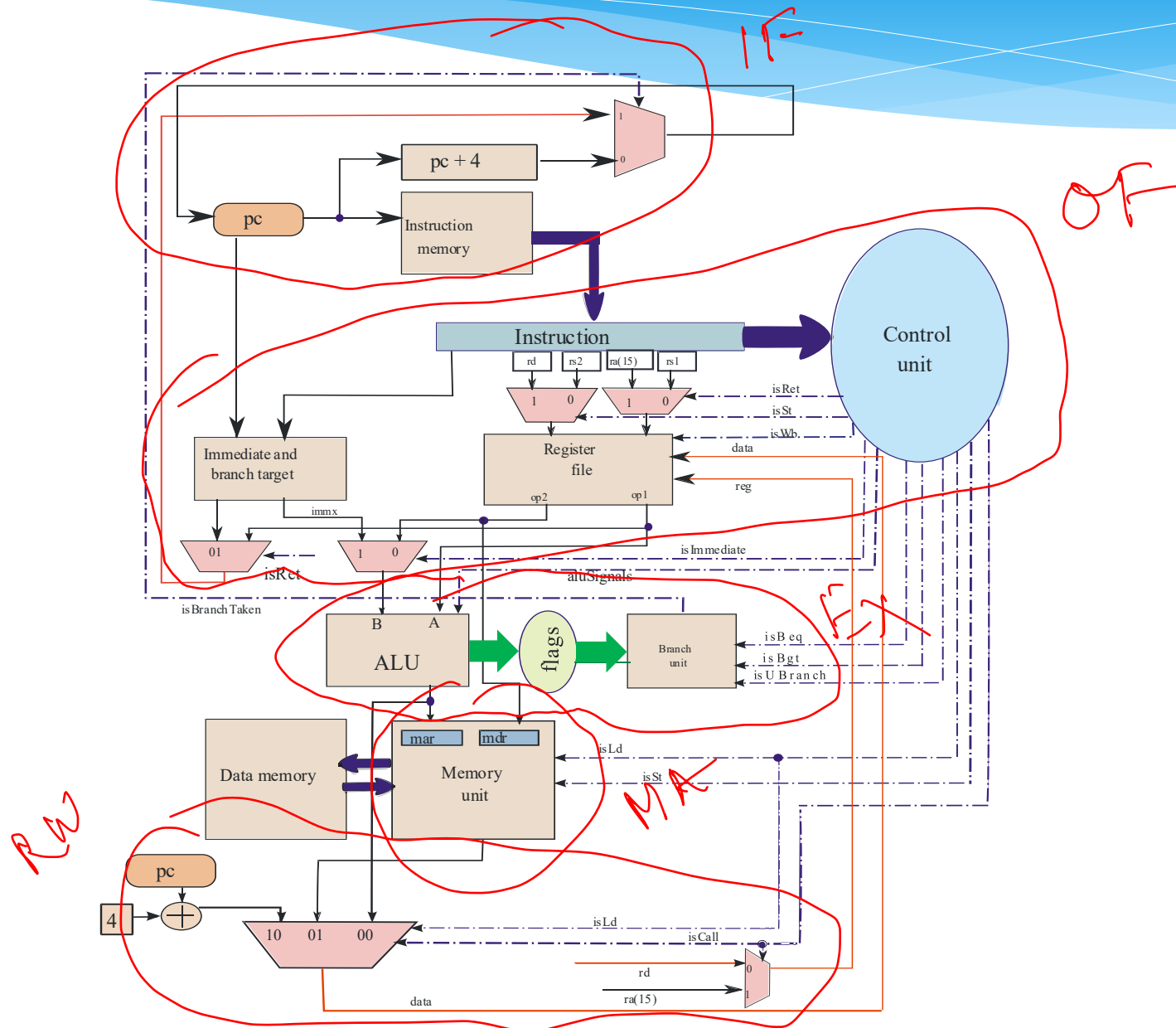


MA Unit



RW Unit



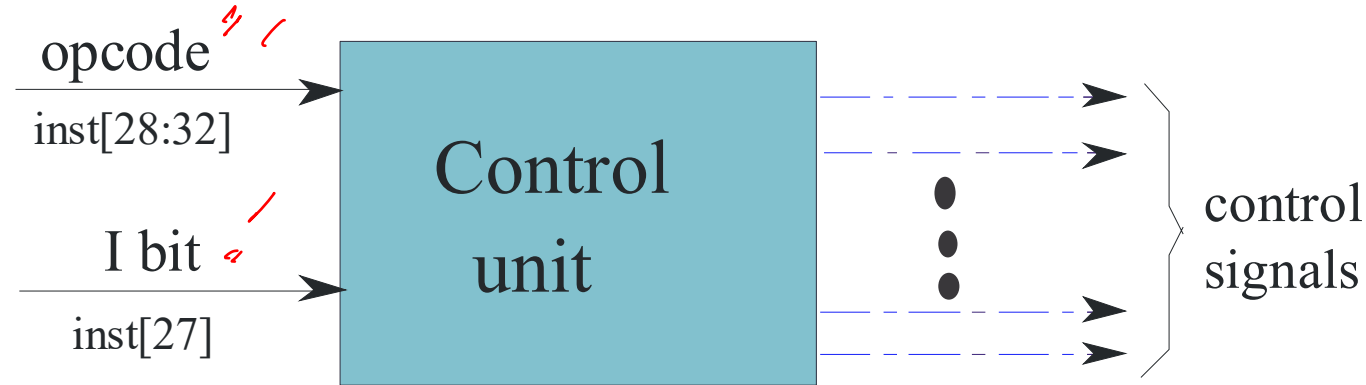


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The Hardwired Control Unit



- * Given the **opcode** and the **immediate** bit
 - * It generates all the **control signals**

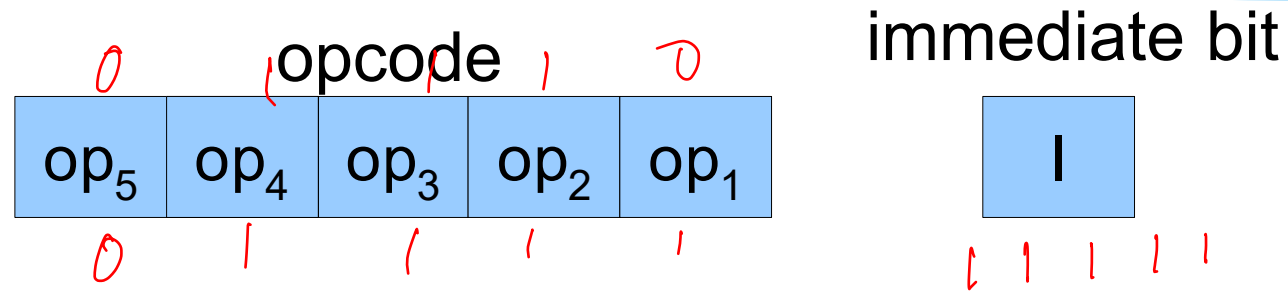
Control Signals

SerialNo.	Signal	Condition
1	<i>isSt</i> ✓	Instruction: <i>st</i>
2	<i>isLd</i> ✓	Instruction: <i>ld</i>
3	<i>isBeq</i> ✓	Instruction: <i>beq</i>
4	<i>isBgt</i> ✓	Instruction: <i>bgt</i>
5	<i>isRet</i> ✓	Instruction: <i>ret</i>
6	<i>isImmediate</i> ✓	<i>I</i> bit set to 1
7	<i>isWb</i> ✓	Instructions: <i>add, sub, mul, div, mod, and, or, not, mov, ld, lsl, lsr, asr, call</i>
8	<i>isUBranch</i> ✓	Instructions: <i>b, call, ret</i>
9	<i>isCall</i> ✓	Instructions: <i>call</i>

Control Signals – II

<i>aluSignal</i>		
10	<i>isAdd</i>	Instructions: <i>add, ld, st</i>
11	<i>isSub</i>	Instruction: <i>sub</i>
12	<i>isCmp</i>	Instruction: <i>cmp</i>
13	<i>isMul</i>	Instruction: <i>mul</i>
14	<i>isDiv</i>	Instruction: <i>div</i>
15	<i>isMod</i>	Instruction: <i>mod</i>
16	<i>isLsl</i>	Instruction: <i>lsl</i>
17	<i>isLsr</i>	Instruction: <i>lsr</i>
18	<i>isAsr</i>	Instruction: <i>asr</i>
19	<i>isOr</i>	Instruction: <i>or</i>
20	<i>isAnd</i>	Instruction: <i>and</i>
21	<i>isNot</i>	Instruction: <i>not</i>
22	<i>isMov</i>	Instruction: <i>mov</i>

Control signal Logic



Serial No.	Signal	Condition
1	<u>isSt</u>	$\overline{op_5} \cdot \overline{op_4} \cdot \overline{op_3} \cdot \overline{op_2} \cdot \overline{op_1}$ ←
2	isLd	$\overline{op_5} \cdot \overline{op_4} \cdot \overline{op_3} \cdot \overline{op_2} \cdot \overline{op_1}$
3	isBeq	$op_5 \cdot \overline{op_4} \cdot \overline{op_3} \cdot \overline{op_2} \cdot \overline{op_1}$
4	isBgt	$op_5 \cdot \overline{op_4} \cdot \overline{op_3} \cdot \overline{op_2} \cdot op_1$
5	isRet	$op_5 \cdot \overline{op_4} \cdot op_3 \cdot \overline{op_2} \cdot \overline{op_1}$
6	isImmediate	I
7	<u>isWb</u>	$\sim(op_5 + \overline{op_5} \cdot op_3 \cdot op_1 \cdot (op_4 + \overline{op_2})) + op_5 \cdot \overline{op_4} \cdot \overline{op_3} \cdot \overline{op_2} \cdot op_1$
8	<u>isUbranch</u>	$op_5 \cdot \overline{op_4} \cdot (\overline{op_3} \cdot op_2 + op_3 \cdot \overline{op_2} \cdot \overline{op_1})$ ✓
9	isCall	$op_5 \cdot \overline{op_4} \cdot \overline{op_3} \cdot op_2 \cdot op_1$

Handwritten notes on the left side:

ALU ——— 00000

CALL ———

LOBB ———

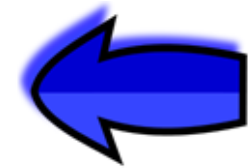
✓ ✓ ✓

Control Signal Logic - II

<i>aluSignals</i>		
10	<i>isAdd</i> ✓	$\overline{op5}.\overline{op4}.\overline{op3}.\overline{op2}.\overline{op1} + \overline{op5}.op4.op3.op2$
11	<i>isSub</i> ✓	$\overline{op5}.\overline{op4}.\overline{op3}.\overline{op2}.op1$
12	<i>isCmp</i> ✓	$\overline{op5}.\overline{op4}.op3.\overline{op2}.op1$
13	<i>isMul</i> ✓	$\overline{op5}.\overline{op4}.\overline{op3}.op2.\overline{op1}$
14	<i>isDiv</i>	$\overline{op5}.\overline{op4}.\overline{op3}.op2.op1$
15	<i>isMod</i>	$\overline{op5}.\overline{op4}.op3.\overline{op2}.\overline{op1}$
16	<i>isLsl</i>	$\overline{op5}.op4.\overline{op3}.op2.\overline{op1}$
17	<i>isLsr</i>	$\overline{op5}.op4.\overline{op3}.op2.op1$
18	<i>isAsr</i>	$\overline{op5}.op4.op3.\overline{op2}.\overline{op1}$
19	<i>isOr</i>	$\overline{op5}.\overline{op4}.op3.op2.op1$
20	<i>isAnd</i>	$\overline{op5}.\overline{op4}.\overline{op3}.op2.\overline{op1}$
21	<i>isNot</i>	$\overline{op5}.op4.\overline{op3}.\overline{op2}.\overline{op1}$
22	<i>isMov</i>	$\overline{op5}.op4.\overline{op3}.\overline{op2}.op1$

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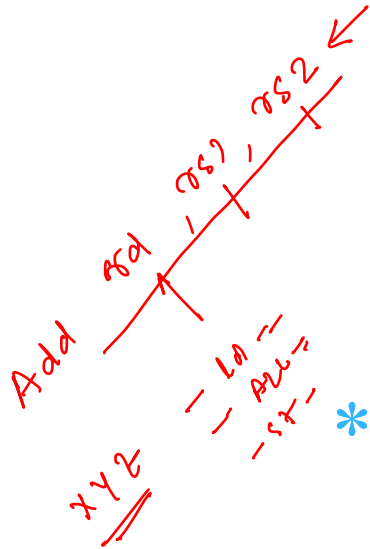
Microprogramming

* Idea of **microprogramming**

- * Expose the elements in a processor to software
- * Implement instructions as dedicated software routines

* Why make the **implementation** of instructions flexible ?

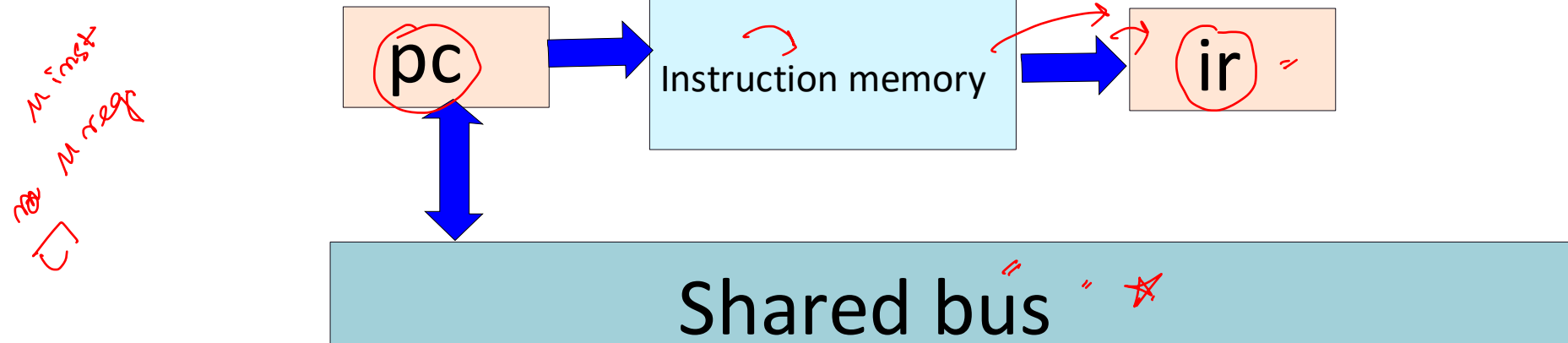
- * Dynamically **change** their behaviour
- * Fix bugs in **implementations**
- * Implement very **complex** instructions



Microprogrammed Data Path

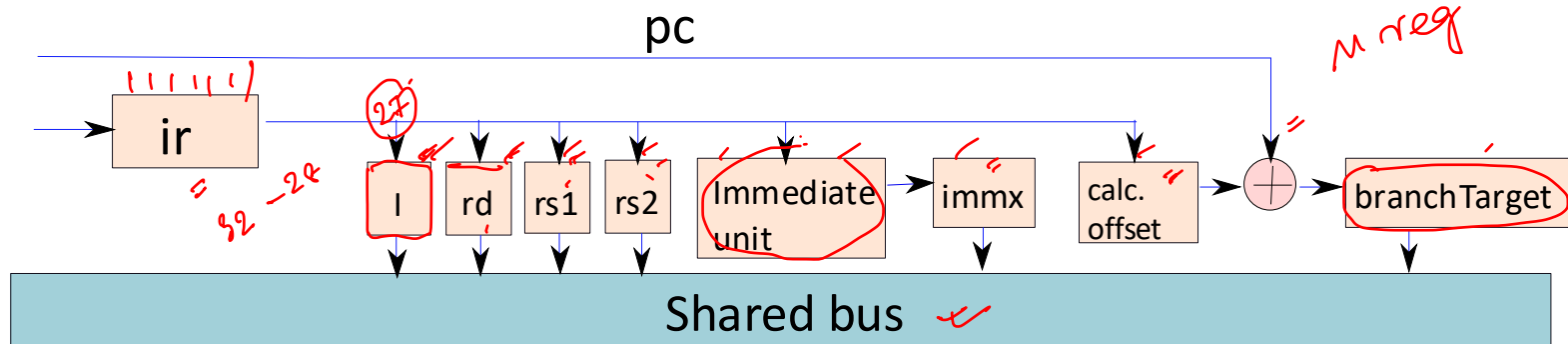
- * Expose all the state elements to dedicated system software – **firmware**
- * Write dedicated routines in **firmware** for implementing each instruction
- * **Basic idea**
 - * 1 SimpleRisc Instruction → Several micro instructions
 - * Execute each micro instruction
 - * We require a microprogram counter, and microinstruction memory

Fetch Unit



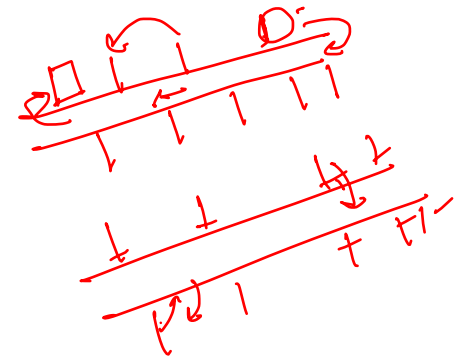
- * The **pc** is used to access the instruction memory.
- * The contents of the instruction are saved in the **instruction register (ir)**

Decode Unit



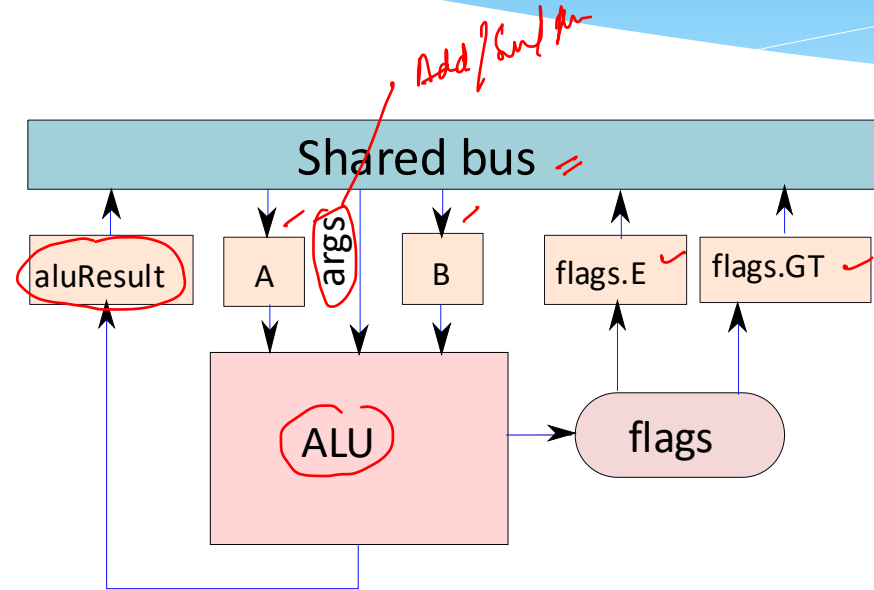
- * Divide the contents of **ir** into different fields

- * **I**, **rd**, **rs1**, **rs2**, **immx**, and **branchTarget**

[illegible]

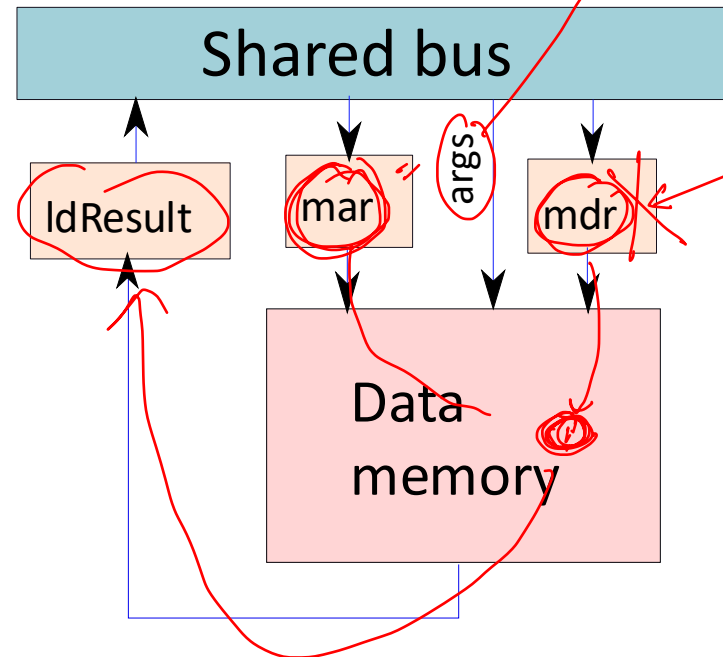
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ALU

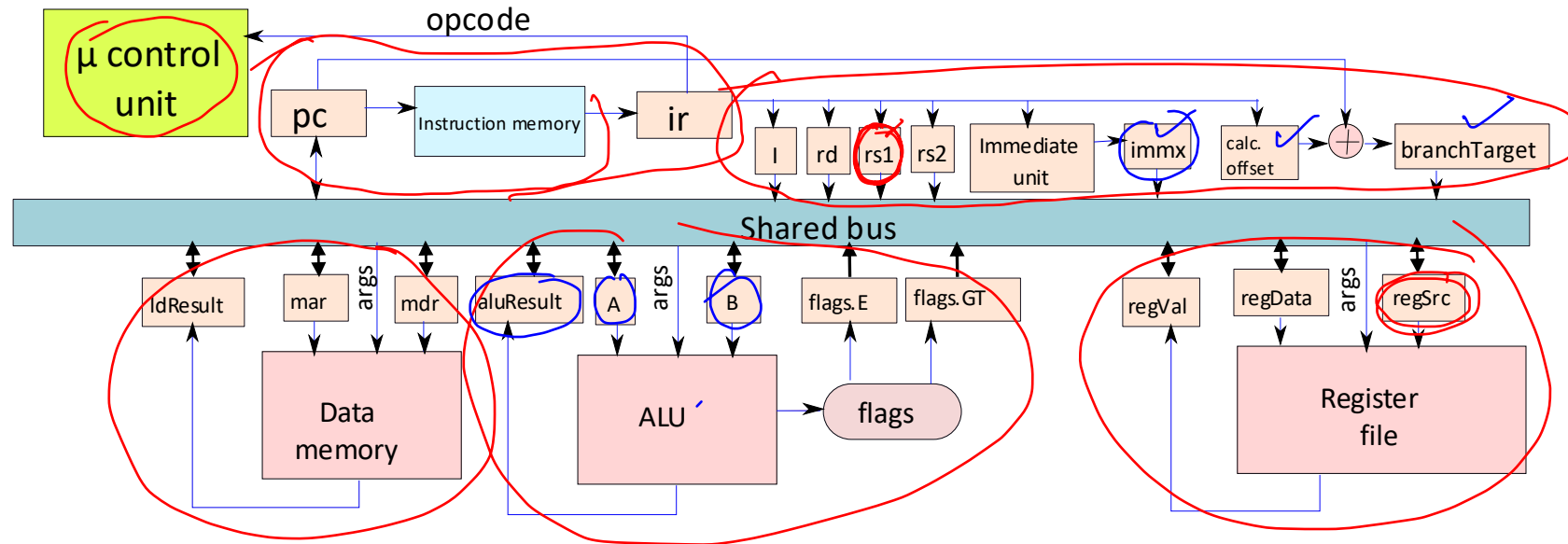


- * A, B → ALU operands
- * args → ALU operation type
- * aluResult → ALU Result

Memory Unit



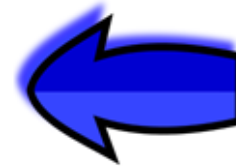
Microprogrammed Data Path



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Add ✓
MOV ✓
CMP ✓



Internal Registers

SerialNo.	Register	Size (bits)	Function
1	<i>pc</i>	32	program counter
2	<i>ir</i>	32	instruction register
3	<i>I</i>	1	immediate bit in the instruction
4	<i>rd</i>	4	destination register id
5	<i>rs 1</i>	4	id of the first source register
6	<i>rs 2</i>	4	id of the second source register
7	<i>immx</i>	32	immediate embedded in the instruction (after processing modifiers)
8	<i>branchTarget</i>	32	branch target, computed as the sum of the PC and the offset embedded in the instruction
9	<i>regSrc</i>	4	contains the id of the register that needs to be accessed in the register file
10	<i>regData</i>	32	contains the data to be written into the register file

Internal Registers - II

11	<i>regVal</i>	32	value read from the register file
12	<i>A</i>	32	first operand of the ALU
13	<i>B</i>	32	second operand of the ALU
14	<i>flags.E</i>	1	the equality flag
15	<i>flags.GT</i>	1	the greater than flag
16	<i>aluResult</i>	32	the ALU result
17	<i>mar</i>	32	memory address register
18	<i>mdr</i>	32	memory data register
19	<i>ldResult</i>	32	the value loaded from memory

Microinstructions

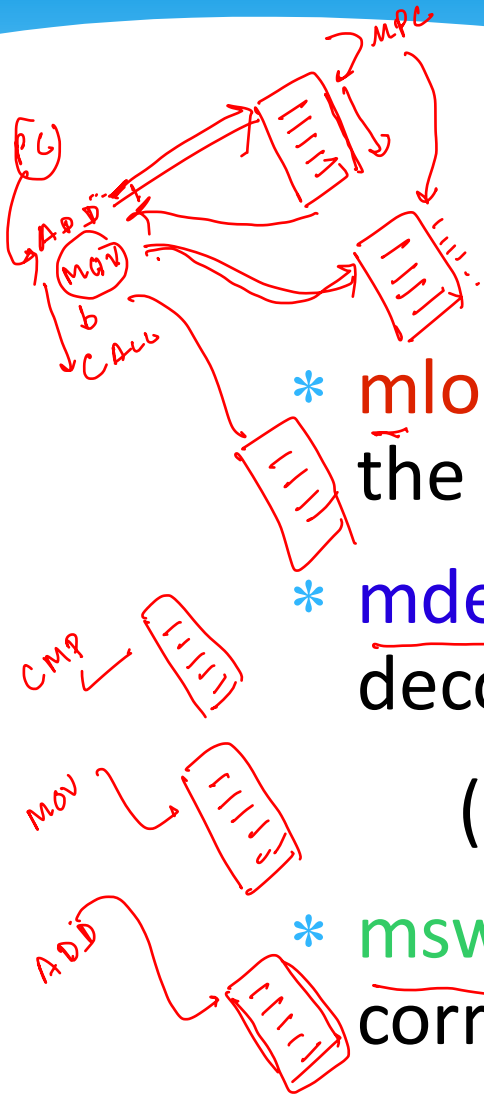
Basic Instructions

* mloadIR → Loads the instruction register (ir) with the contents of the instruction. = IF

* mdecode → Waits for 1 cycle. Meanwhile, all the decode registers get populated

(I, Id, rs1, rs2, immx, branchTarget)

* mswitch → Loads the set of micro instructions corresponding to a program instruction.



Move Microinstructions

- * mmov r1, r2 : $r1 \leftarrow r2$
- * mmov r1, r2, <args> : $r1 \leftarrow r2$, send the value of args on the bus
READ/WRITE
- * mmovi r1, <imm> : $r1 \leftarrow \text{imm}$

Add and Branch Microinstructions

* madd r1, imm, <args>

* $r1 \leftarrow r1 + \text{imm}$

* send <args> on the bus

* mbeq r1, imm, <label>

* if ($r1 == \text{imm}$), $\mu pc = \text{addr}(\text{label})$

* mb <label>

* $\mu pc = \text{addr}(\text{label})$

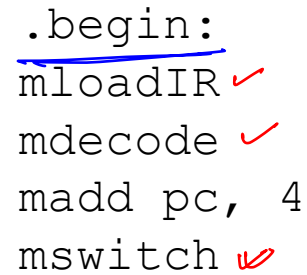
Summary of Microinstructions

SerialNo.	Microinstruction	Semantics
1	<i>mloadIR</i> ✓	$ir \leftarrow [pc]$
2	<i>mdecode</i> ✓	populate all the decode registers
3	<i>mswitch</i> ✓	jump to the μpc corresponding to the opcode
4	<i>mmov reg1, reg2, < <u>args</u> ></i>	$reg1 \leftarrow reg2$, send the value of <i>args</i> to the unit that owns <i>reg1</i> , <i>< args ></i> is optional
5	<i>mmovi reg1, imm, < <u>args</u> ></i>	$reg1 \leftarrow imm$, <i>< args ></i> is optional
6	<i>madd reg1, imm, < args ></i>	$reg1 \leftarrow reg1 + imm$, <i>< args ></i> is optional
7	<i>mbeq reg1, imm, < label ></i>	if ($reg1 = imm$) $\mu pc \leftarrow addr(label)$
8	<i>mb <label></i>	$\mu pc \leftarrow addr(label)$

Implementing Instructions in Microcode

* The microcode preamble

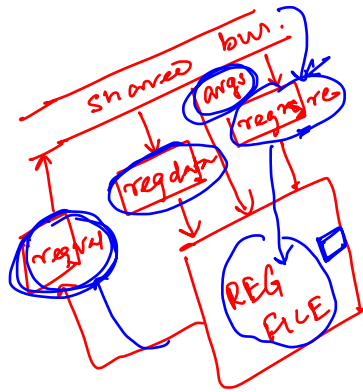
```
.begin:  
mloadIR ✓  
mdecode ✓  
madd pc, 4  
mswitch ✓
```



The diagram shows a box containing the microcode preamble instructions. To the right of the instructions, there are two vertical red lines. Above the box, there is a red circle with 'ADD' and 'MOV' written inside, with an arrow pointing to a red box containing five horizontal lines. An arrow also points from the red box to the label 'MPC'.

- * Load the program counter
- * Decode the instruction
- * Add 4 to the pc
- * Switch to the first microinstruction in the microcode sequence of the prog. instruction

3 Address Format ALU Instruction



ALU

ADD rd, rs1, rs2
ADD rd, rs1, imm

```

/* transfer the first operand to the ALU */
mmov regSrc, rs1, <read>
mmov A, regVal

/* check the value of the immediate register */
mbeq I, 1, .imm X
/* second operand is a register */
mmov regSrc, rs2, <read>
mmov B, regVal, <aluop> ADD
mb .rw

/* second operand is an immediate */
.imm:
mmov B, immx, <aluop> ADD

/* write the ALU result to the register file*/
.rw:
mmov regSrc, rd
mmov regData, aluResult, <write>
mb .begin
    
```

ALU result

MOV rd rs2/imm.

mbeq I, 1, .imm

mMOV regse, rs2, <read> ← regval is populated

mMOV regdata, regval -
mb .rw

• imm
mMOV regdata, immx -

• rw
mMOV regse, rd, <write>

mb -begin.

The mov Instruction

—mov instruction—

```
/* check the value of the immediate register */
mbeq I, 1, .imm
/* second operand is a register */
mmov regSrc, rs2, <read>
mmov regData, regVal
mb .rw
/* second operand is an immediate */
.imm:
mmov regData, immx

/* write to the register file*/
.rw:
mmov regSrc, rd, <write>

/* jump to the beginning */
mb .begin
```

The *not* Instruction

— *not* instruction —

```
/* check the value of the immediate register */
mbeq I, 1, .imm
/* second operand is a register */
mmov regSrc, rs2, <read>
mmov B, regVal, <not> /* ALU operation */
mb .rw
/* second operand is an immediate */
.imm:
mmov B, immx, <not> /* ALU operation */

/* write to the register file*/
.rw:
mmov regData, aluResult
mmov regSrc, rd, <write>

/* jump to the beginning */
mb .begin
```

The *cmp* Instruction

—*cmp* instruction—

```
/* transfer rs1 to register A */  
mov regSrc, rs1, <read>  
mov A, regVal  
  
/* check the value of the immediate register  
mbeq I, 1, .imm  
  
/* second operand is a register */  
mmov regSrc, rs2, <read>  
mmov B, regVal, <cmp> /* ALU operation */  
mb .begin  
  
/* second operand is an immediate */  
.imm:  
mmov B, immx, <cmp> /* ALU operation */  
mb .begin
```

The *nop* Instruction

- * `mb .begin`

The *ld* Instruction

—— *ld* instruction ——

```
/* transfer rs1 to register A */  
mmov regSrc, rs1, <read>  
mmov A, regVal  
  
/* calculate the effective address */  
mmov B, immx, <add> /* ALU operation */  
  
/* perform the load */  
mmov mar, aluResult, <load>  
  
/* write the loaded value to the register file */  
mmov regData, ldResult  
mmov regSrc, rd, <write>  
  
/* jump to the beginning */  
mb .begin
```

The *st* Instruction

— *st* instruction —

```
/* transfer rsl to register A */  
mmov regSrc, rsl, <read>  
mmov A, regVal  
  
/* calculate the effective address */  
mmov B, immx, <add> /* ALU operation */  
  
/* perform the store */  
mmov mar, aluResult  
mmov regSrc, rd, <read>  
mmov mdr, regVal, <store>  
  
/* jump to the beginning */  
mb .begin
```

beq and *bgt* Instructions

—*beq* instruction—

```
/* test the flags register  
mbeq flags.E, 1, .branch  
mb .begin
```

```
.branch:  
mmov pc, branchTarget  
mb .begin
```

—*bgt* instruction—

```
/* test the flags register  
mbeq flags.GT, 1, .branch  
mb .begin
```

```
.branch:  
mmov pc, branchTarget  
mb .begin
```

call Instruction

call instruction

```
/* save PC + 4 in the return address register */  
mmov regData, pc  
mmovi regSrc, 15, <write>  
  
/* branch to the function */  
mmov pc, branchTarget  
mb .begin
```

ret Instruction

ret instruction

```
/* save the contents of the return  
   address register in the PC */
```

```
mmovi regSrc, 15, <read>  
mmov pc, regVal  
mb .begin
```

Example

Change the call instruction to store the return address on the stack. The preamble need not be shown.

Answer:

```
——— stack based call instruction ———
/* read the stack pointer */
mmovi regSrc, 14, <read>
madd regVal, -4 /* decrement the stack pointer */

/* set the memory address to the stack pointer */
mmov mar, regVal

/* update the stack pointer */
mmov regData, regVal, <write> /* update stack pointer */

/* write the return address to the stack */
mmov mdr, pc, <store>

/* jump to the beginning */
mb .begin
```